Architecture overview

[redacted]

A Recurrent Neural Network (RNN) is a type of neural network designed to process **sequential data** by maintaining a **hidden state (memory)** that carries information from previous time steps. Unlike traditional feedforward networks, RNNs use **recurrent connections (loops)**, allowing them to capture **temporal dependencies and context** in data.

This makes RNNs especially useful for tasks where **order matters**, such as time series prediction, speech recognition, and natural language processing.

Example sentence:

"Hi this is Akash, I am an instructor at Phitron"

RNN reads **one word at a time (time steps)** and updates its **hidden state (memory)**.

RNN Processing Table

Time Step (t)	Input Word	Previous Hidden State (h_{t-1})	Current Hidden State (h_t)	What it "remembers"
1	Hi	0 (initial)	h_1	Greeting started
2	this	h_1	h_2	Intro sentence forming
3	is	h_2	h_3	Sentence structure building
4	Akash	h_3	h_4	Learns **name = Akash**
5	,	h_4	h_5	Pause / separation
6	I	h_5	h_6	New clause starts
7	am	h_6	h_7	Self-description
8	an	h_7	h_8	Continuing phrase
9	instructor	h_8	h_9	Learns **profession = instructor**
10	at	h_9	h_{10}	Linking info
11	Phitron	h_{10}	h_{11}	Learns **organization = Phitron**

- So it **combines current word + previous memory**.

Final Understanding

By the end (h_{11}), the RNN has captured:

- Name → Akash
- Role → Instructor
- Organization → Phitron